Intelligent Reflecting Surface Enhanced Multi-User MISO Symbiotic Radio Systems

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***Abstract*—To support massive access for future wireless communications, we propose a novel intelligent reflecting surface (IRS) enhanced downlink multi-user multi-input single-output (MU-MISO) symbiotic radio (SR) system, where each IRS, acting as a reflecting Internet-of-Things (IoT) device, transmits its message to a nearby primary receiver (PR) by reflecting the RF signals from the primary transmitter (PT), and simultaneously enhances the transmission from the PT to the associated PR. Thus, each PR jointly decodes its own message as well as the one from the corresponding IRS. We are interested in maximizing the weighted sum-rate of both primary and IoT transmissions by jointly designing the active transmit beamforming at PT and the passive beamforming at each IRS, subject to the maximum transmit power constraint at PT. Besides, as the passive elements at IRS can only reflect the incident signal with discrete phase shifts in practice, the discrete reflection coefficient (RC) constraint is further considered at the IRSs. Due to the non-convexity of the formulated problems, we solve them with fractional programming (FP) technique and alternating optimization (AO) method. Simulation results have verified the effectiveness of the proposed algorithms compared to different benchmark schemes.**

I. Introduction

With the widespread deployment of Internet-of-Things (IoT) networks, the number of IoT devices that need to be supported per square meter may reach 100 in the near future [1]. This introduces great challenges to the network operations due to the limited energy and spectrum resources [2], [3]. Recently, symbiotic radio (SR) [4]–[7] has been proposed as a promising solution to overcome these challenges. In particular, SR enables IoT devices, also known as backscatter devices (BDs), to transmit information to nearby receivers over the ambient primary RF carriers via backscatter modulation. Besides, by introducing cooperation between the primary and IoT transmissions, SR not only supports the IoT communication over the primary signals, but also enhances the performance of the primary transmission. Due to the spectrum, power, and infrastructure sharing nature, symbiotic radio has become a spectrum-, energy- and cost-efficient solution for future IoT communications.

Specifically, in [4], the ergodic rate and the outage probability have been investigated for the symbiotic cellular and IoT networks. In [5], the resource allocation and transceiver design have been studied to enhance the performance of the SR system. In the above literature, the IoT devices are equipped with a single antenna. In practice, because of the double fading effect, the backscatter link in SR is usually much weaker than the direct link, thus the performance gain obtained by the primary system through cooperation may not be significant [6].

Recently, intelligent reflecting surface (IRS) has been introduced into SR systems to enhance the backscatter links in [6]. Unlike the single-antenna BD, IRS is an artificial passive radio structure consisting of multiple reflection elements, which can reflect the incident RF waves into specific directions with low power consumption [8]. It is shown that in [6], both primary and IoT transmissions can achieve significant performance improvement by passive beamforming due to the enhanced backscatter link. Motivated by the above results, in this paper, we study a novel IRS enhanced downlink multi-user multi-input single-output (MU-MISO) SR system, as shown in Fig. 1. In particular, each IRS is applied to assist the primary transmission from the PT to one of the nearby PRs, and at the same time, it also transmits its own messages to the same PR.

The proposed IRS enhanced MU-MISO SR system describes a practical scenario where each IRS is embedded with multiple sensors in addition to reflecting the incident signals. Thus each IRS can monitor the surrounding environment in terms of temperature, humidity, acoustic noise, etc., and feeds the collected location-specific environmental data to the nearby PR of interest. Therefore, with the proposed system, the PRs can not only benefit from the enhanced primary communication due to the enhanced backscatter links, but also receive useful local information about the surrounding environment in real time.

The main contributions of this paper are summarized as follows:

1) To the best of our knowledge, this paper is the first to study the IRS enhanced MU-MISO SR system, which can achieve enhanced primary transmission with the aid of IRSs, and at the same time enable the primary receivers to receive useful local information from nearby IRS via IoT transmission.

2) The joint active beamforming at PT and passive beamforming at each IRS design problem is formulated to maximize the weighted sum-rate (WSR) of both primary and IoT transmissions with the maximum transmit power constraints at the PT and various practical constraints on the RCs. Since the formulated problem is non-convex, an effective algorithm based on alternating optimization (AO) [9] and fractional programming (FP) [10] is proposed to solve it.

3) Simulation results show that, the IRSs in the MU-MISO SR system can not only enhance the primary transmission as compared to different baseline systems, but also transmit

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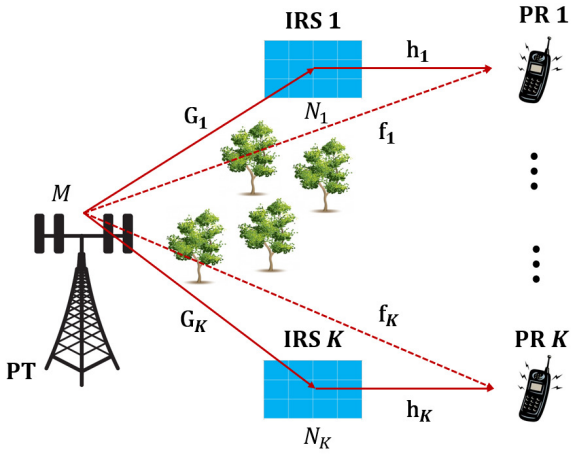

Fig. 1: IRS enhanced MU-MISO SR communication system

their own messages to the nearby receivers. Besides, passive beamforming based on 3-bit phase-shifters can achieve very close performances to those of the ideal cases.

It is pointed out that the IRS has been applied to conventional communications for the purpose of enhancing the transmission performance only. For example, the received signal power maximization problem has been studied in [11] by jointly designing the active and passive beamforming, and the WSR maximization problem has been studied in [12] by jointly designing beamformers and artificial noise. By further considering the discrete reflection coefficient (RC) values at the IRS due to the hardware constraint, a parallel iterative algorithm has been proposed to solve the WSR maximization problem in [9]. Unlike the works mentioned above, the IRS in our proposed SR system, can help to enable simultaneous IoT transmissions through the spectrum, power, and infrastructure sharing in addition to enhancing the performance of the primary system.

II. System model and Problem Formulation

A. System model

As shown in Fig. 1, we consider an IRS enhanced downlink MU-MISO SR system, consisting of a PT with M antennas, K PRs with a single antenna, and K IRSs with multiple reflection elements. Each IRS is associated with one nearby PR to support the primary communication from PT to this PR, and simultaneously, it also transmits its information, such as the sensor data, to the associated PR. More specifically, each IRS modulates its information over the incident signals from the PT and reflects them to the associated PR. This allows the IRSs to transmit their own information without any dedicated RF sources. For simplicity, we let IRS k denote the IRS that communicates with PR k.

In this paper, we consider a block-fading channel model. The channel from the PT to PR k, from PT to IRS k, from IRS k to PR k, are denoted by $\mathbf{f}_k \in \mathbb{C}^{M\times 1}$, $\mathbf{G}_k \in \mathbb{C}^{N_k\times M}$ and $\mathbf{h}_k \in \mathbb{C}^{N_k\times 1}$, respectively, where N_k is the number of reflection elements for IRS k. The phase-shift matrix of IRS k is denoted by $\mathbf{\Phi}_k = |\alpha_k|\text{diag}(\theta_{k1}, \theta_{k2}, \cdots, \theta_{kN_k})$, where $|\alpha_k| \leq 1$ indicates the reflection efficiency of IRS k, and θ_{kn} denotes the RC of the n-th reflection element on IRS k. We consider three different sets of RCs:

$$\mathcal{F}_1 = \{\theta_{kn} | \|\theta_{kn}\|^2 \leqq 1\}, \tag{1}$$

$$\mathcal{F}_2 = \{\theta_{kn} | \theta_{kn} = e^{j\varphi_{kn}}, \varphi_{kn} \in [0, 2\pi]\}, \tag{2}$$

$$\mathcal{F}_3 = \left\{\theta_{kn} = e^{j\varphi_{kn}} | \varphi_{kn} \in \left\{0, \frac{2\pi}{\tau}, \cdots, \frac{2\pi(\tau-1)}{\tau}, \tau = 2^m\right\}\right\}. \tag{3}$$

B. Received Signals at PR k

Let c_k be the signal transmitted to PR k by IRS k, and we assume that c_k is binary phase shift keying (BPSK) symbol with a period T_c, i.e., $c_k = \{-1, 1\}$. Furthermore, we assume that the PT transmits N_T primary symbols during each symbol period of the IRS, i.e., $T_c = N_T T_s$, where T_s is the symbol period of the primary signal from the PT. Denote $s_k(n)$ as the n-th signal from the PT to PR k during a IRS's symbol period. We assume that $s_k(n)$ is independent random variable and $s_k(n) \sim \mathcal{CN}(0, 1)$. Thus, the n-th received signal at PR k can be expressed as

$$y_k(n) = (\mathbf{f}_k + c_k \mathbf{b}_k)^H \sum_{i=1}^{K} \mathbf{v}_i s_i(n) + u_k, \tag{4}$$

where $\mathbf{b}_k = \mathbf{G}_k^H \mathbf{\Phi}_k \mathbf{h}_k$ denotes the backscatter link of IRS k, $\mathbf{v}_i \in \mathbb{C}^{M\times 1}$ denotes the transmit beamforming vector for PR k, and $u_k \sim \mathcal{CN}(0, \sigma_k^2)$ denotes the received noise at PR k.

For the received signal of PR k (4), we don't consider the received signals from all IRSs but only the signal from IRS k. This is because we assume that each PR is close to its associated IRS but far away from the other IRSs. As the reflected signals from them are significantly attenuated by obstacles such as buildings and trees due to high penetration and diffraction losses, thus the direct links between PR and other IRSs can be assumed to be blocked.

It is seen that $s_i(n)$ and c_k are coupling in the second term of (4). In this case, with the maximum likelihood (ML) receiver proposed in [13], $s_i(n)$ and c_k can be decoded jointly to achieve a better performance. The specific process is shown as follows.

According to [13], when PR k decodes $s_i(n)$ and c_k jointly, the backscatter link can be treated as a multipath component from PT to PR k. Therefore, the decoding signal-to-interference-plus-noise ratio (SINR) of $s_k(n)$ at PR k is

$$\gamma_{s,k} = \frac{\left|(\mathbf{f}_k + c_k \mathbf{b}_k)^H \mathbf{v}_k\right|^2}{\sum\limits_{i=1, i\neq k}^{K} \left|(\mathbf{f}_k + c_k \mathbf{b}_k)^H \mathbf{v}_i\right|^2 + \sigma_k^2}, \tag{5}$$

The expression of $\gamma_{s,k}$ contains c_k, which changes relatively fast in contrast to the channel variation. In this case, according

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to [14], we should take expectation over c_k to calculate the achievable rate of $s_i(n)$, which is given by

$$R_{\mathrm{s},k} = \mathbb{E}_{c_k}\{\log_2(1+\gamma_{\mathrm{s},k})\}. \tag{6}$$

According to [13], when $N_T \gg 1$, the SINR for decoding c_k via maximal ratio combining (MRC) at PR k is given by

$$\gamma_{\mathrm{c},k} = \frac{N_T|\mathbf{b}_k\mathbf{v}_k|^2}{\sum\limits_{i=1,i\neq k}^{K}\left|\mathbf{f}_k^H\mathbf{v}_i\right|^2 + \sum\limits_{i=1,i\neq k}^{K}\left|\mathbf{b}_k^H\mathbf{v}_i\right|^2 + \sigma_k^2}. \tag{7}$$

Therefore, the IoT rate of IRS k is given by

$$R_{\mathrm{c},k} = \frac{1}{N_T}\log_2(1+\frac{1}{\Gamma}\gamma_{\mathrm{c},k}). \tag{8}$$

where Γ is the SINR gap to capacity, and it gives an approximate relation between the SINR and the rate [15].

C. Problem Formulation

In this paper, we aim to maximize the weighted sum-rate of all the $s_k(n)$ and c_k for $k = 1, 2, ..., K$ by jointly designing the transmit beamforming at the PT and the passive beamforming at each IRS. Let $\mathbf{W} = [\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_K] \in \mathbb{C}^{M\times K}$ and $\mathbf{\Theta} = [\mathbf{\Phi}_1, \mathbf{\Phi}_2, ..., \mathbf{\Phi}_k]$. The WSR maximization problem can be formulated as

$$\text{(P1):} \quad \max_{\mathbf{W},\mathbf{\Theta}} \quad f(\mathbf{W},\mathbf{\Theta}) = \sum_{k=1}^{K}\rho_{\mathrm{s},k}R_{sk} + \sum_{k=1}^{K}\rho_{\mathrm{c},k}R_{ck}$$

$$\text{s.t.} \quad \theta_{kn} \in \mathcal{F}_i \ (i \in \{1,2,3\}), \tag{9}$$

$$\sum\nolimits_{k=1}^{K}\|\mathbf{v}_k\|_2^2 \le P_m, \tag{10}$$

where $\rho_{\mathrm{s},k}$ and $\rho_{\mathrm{c},k}$ are the weight factors for $R_{\mathrm{s},k}$ and $R_{\mathrm{c},k}$, respectively, and P_m denotes the maximum transmit power at PT. Problem (P1) is very challenging to solve due to the non-convex objective function $f(\mathbf{W},\mathbf{\Theta})$ and the non-convex constraints under $\mathcal{F}_2$ and $\mathcal{F}_3$. Therefore, we aim to find a suboptimal algorithm for (P1).

III. Proposed solutions

In this section, we adopt the method of FP and AO to solve (P1). To be specific, with FP, the objective function is reformulated to a form that is suitable for the subsequent AO first, and then AO is adopted to optimize $\mathbf{W}$ and $\mathbf{\Theta}$ separately.

First, as c_k is BPSK symbol, the mathematical expectation involved in $f(\mathbf{W},\mathbf{\Theta})$ is replaced with the corresponding sample mean value. Next, we apply the Lagrangian dual reformulation proposed in [10] to tackle the logarithm operation. With these two techniques, (P1) can be approximated as

$$\text{(P1-a):} \quad \max_{\mathbf{W},\mathbf{\Theta},\boldsymbol{\beta}} \quad f_r(\mathbf{W},\mathbf{\Theta},\boldsymbol{\beta})$$

$$\text{s.t.} \quad (10).$$

The new objective function is defined by

$$\begin{aligned} &f_r(\mathbf{W},\mathbf{\Theta},\boldsymbol{\beta}) \\ &= \sum_{k=1}^{K}\sum_{j=1}^{2}\frac{\rho_{\mathrm{s},k}}{2}\log_2(1+\beta_{\mathrm{s},kj}) - \sum_{k=1}^{K}\sum_{j=1}^{2}\frac{\rho_{\mathrm{s},k}}{2}\beta_{\mathrm{s},kj} \\ &+ \sum_{k=1}^{K}\sum_{j=1}^{2}\frac{\rho_{\mathrm{s},k}}{2}(1+\beta_{\mathrm{s},kj})\frac{\left|(\mathbf{f}_k+\tilde{c}_j\mathbf{b}_k)^H\mathbf{v}_k\right|^2}{\left|(\mathbf{f}_k+\tilde{c}_j\mathbf{b}_k)^H\mathbf{v}_k\right|^2+\sigma_{\mathrm{s},kj}^2} \\ &+ \sum_{k=1}^{K}\frac{\rho_{\mathrm{c},k}}{N_T}\log_2(1+\beta_{\mathrm{c},k}) - \sum_{k=1}^{K}\frac{\rho_{\mathrm{c},k}}{N_T}\beta_{\mathrm{c},k} \\ &+ \sum_{k=1}^{K}\frac{\rho_{\mathrm{c},k}}{\Gamma}(1+\beta_{\mathrm{c},k})\frac{\left|\mathbf{b}_k^H\mathbf{v}_k\right|^2}{(N_T/\Gamma)\left|\mathbf{b}_k^H\mathbf{v}_k\right|^2+\sigma_{\mathrm{c},k}^2}, \end{aligned} \tag{11}$$

where

$$\sigma_{\mathrm{s},kj}^2 = \sum\nolimits_{i=1,i\neq k}^{K}\left|(\mathbf{f}_k+\tilde{c}_j\mathbf{b}_k)^H\mathbf{v}_i\right|^2+\sigma_k^2,$$

$$\sigma_{\mathrm{c},k}^2 = \sum\nolimits_{i=1,i\neq k}^{K}\left|\mathbf{f}_k^H\mathbf{v}_i\right|^2+\sum\nolimits_{i=1,i\neq k}^{K}\left|\mathbf{b}_k^H\mathbf{v}_i\right|^2+\sigma_k^2,$$

$\boldsymbol{\beta}$ is a collection of auxiliary variables including β_{skj} and β_{ck} ($k \in \{1,2,...,K\}, j \in \{1,2\}$), and $\tilde{c}_1 = 1$, $\tilde{c}_2 = -1$.

For given value of $\mathbf{W}$ and $\mathbf{\Theta}$, it can be shown easily that (P1-a) is a convex problem with respect to β_i ($\beta_{\mathrm{s},kj}$ or $\beta_{\mathrm{c},k}$), the optimal value of which can be obtained as follows

$$\beta_{\mathrm{c},k}^o = \frac{1}{\Gamma}\frac{N_T\left|\mathbf{b}_k^H\mathbf{v}_k\right|^2}{\sigma_{\mathrm{c},k}^2}, \tag{12}$$

$$\beta_{\mathrm{s},kj}^o = \frac{\left|(\mathbf{f}_k+\tilde{c}_j\mathbf{b}_k)^H\mathbf{v}_k\right|^2}{\sigma_{\mathrm{s},kj}^2}. \tag{13}$$

by setting $\partial f_r(\mathbf{W},\mathbf{\Theta},\boldsymbol{\beta})/\partial\beta_{\mathrm{c},k}$ and $\partial f_r(\mathbf{W},\mathbf{\Theta},\boldsymbol{\beta})/\partial\beta_{\mathrm{s},kj}$ to zero. Next, we will apply the AO method to separately optimize $\mathbf{W}$ and $\mathbf{\Theta}$.

A. Optimize $\boldsymbol{W}$

In this subsection, we consider the optimization of $\mathbf{W}$ for given value of $\boldsymbol{\beta}$ and $\mathbf{\Theta}$. As $\mathbf{W}$ are only involved in the third and the last ratio terms of (11), which cannot be optimized directly, we apply the quadratic transform [10] to transform the fractional forms. As a result, $f_r(\mathbf{W})$ to

$$\begin{aligned} &f_{q1}(\mathbf{W},\boldsymbol{\xi}) \\ &= \sum_{k=1}^{K}\sum_{j=1}^{2}[2\,\mathrm{Re}\{\xi_{\mathrm{s},kj}^*\sqrt{\frac{\rho_{\mathrm{s},k}}{2}(1+\beta_{\mathrm{s},kj})}(\mathbf{f}_k+\tilde{c}_j\mathbf{b}_k)^H\mathbf{v}_k)\} \\ &- |\xi_{\mathrm{s},kj}|^2(\left|(\mathbf{f}_k+\tilde{c}_j\mathbf{b}_k)^H\mathbf{v}_k\right|^2+\sigma_{\mathrm{s},kj}^2)] \\ &+ \sum_{k=1}^{K}[2\,\mathrm{Re}\{\xi_{\mathrm{c},k}^*\sqrt{\frac{\rho_{\mathrm{c},k}}{\Gamma}(1+\beta_{\mathrm{c},k})}\mathbf{b}_k^H\mathbf{v}_k\} \end{aligned} \tag{14}$$

$$- |\xi_{\mathrm{c},k}|^2(\frac{N_T}{\Gamma}\left|\mathbf{b}_k^H\mathbf{v}_k\right|^2+\sigma_{\mathrm{c},k}^2)] + \mathrm{const}(\boldsymbol{\beta}), \tag{15}$$

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where $\boldsymbol{\xi}$ is a set of new auxiliary variables, and const($\boldsymbol{\beta}$) denotes a constant term when $\boldsymbol{\beta}$ is fixed. After the quadratic transform, the equivalent optimization problem becomes

$$\text{(P1-b):} \quad \max_{\mathbf{W},\boldsymbol{\xi}} \quad f_{q1}(\mathbf{W},\boldsymbol{\xi}) \quad \text{s.t.} \quad (10).$$

Obviously, (P1-b) is a convex problem with respect to ξ_i ($\xi_{\mathrm{s},kj}$ or $\xi_{\mathrm{c},k}$) or $\mathbf{v}_k$. Thus, AO method is applied to optimize $\boldsymbol{\xi}$ and $\mathbf{W}$ alternatively. The optimal solution is expressed as:

$$\xi_{\mathrm{c},k}^o = \sqrt{\frac{\rho_{\mathrm{c},k}}{\Gamma}(1+\beta_{\mathrm{c},k})}\frac{\mathbf{b}_k^H\mathbf{v}_k}{(N_T/\Gamma)\left|\mathbf{b}_k^H\mathbf{v}_k\right|^2+\sigma_{\mathrm{c},k}^2}, \tag{16}$$

$$\xi_{\mathrm{s},kj}^o = \sqrt{\frac{\rho_{\mathrm{s},k}}{2}(1+\beta_{\mathrm{s},kj})}\frac{(\mathbf{f}_k+\tilde{c}_j\mathbf{b}_k)^H\mathbf{v}_k}{\left|(\mathbf{f}_k+\tilde{c}_j\mathbf{b}_k)^H\mathbf{v}_k\right|^2+\sigma_{\mathrm{s},kj}^2}, \tag{17}$$

and

$$\mathbf{v}_k^o = (\lambda_0 I_M + \mathbf{B}_{\mathrm{v},k})^{-1}\mathbf{a}_{\mathrm{v},k}, \tag{18}$$

where

$$\mathbf{a}_{\mathrm{v},k} = \sqrt{\frac{\rho_{\mathrm{c},k}}{\Gamma}(1+\beta_{\mathrm{s},kj})}\xi_{\mathrm{c},k}\mathbf{b}_k + \sum_{j=1}^{2}\sqrt{\frac{\rho_{\mathrm{s},k}}{2}(1+\beta_{\mathrm{s},kj})}\xi_{\mathrm{s},kj}(\mathbf{f}_k+\tilde{c}_j\mathbf{b}_k), \tag{19}$$

$$\begin{aligned}\mathbf{B}_{\mathrm{v},k} =& \sum_{j=1}^{2}|\xi_{\mathrm{s},kj}|^2(\mathbf{f}_k+\tilde{c}_j\mathbf{b}_k)(\mathbf{f}_k+\tilde{c}_j\mathbf{b}_k)^H \\ &+\sum_{j=1}^{2}\sum_{i=1,i\neq k}^{K}|\xi_{\mathrm{s},ij}|^2(\mathbf{f}_i+\tilde{c}_j\mathbf{b}_i)(\mathbf{f}_i+\tilde{c}_j\mathbf{b}_i)^H \\ &+|\xi_{\mathrm{c},k}|^2(\frac{N_T}{\Gamma}\mathbf{b}_k\mathbf{b}_k^H)+\sum_{i=1,i\neq k}^{K}|\xi_{\mathrm{c},i}|^2(\mathbf{f}_i\mathbf{f}_i^H+\mathbf{b}_i\mathbf{b}_i^H).\end{aligned} \tag{20}$$

In (18), λ_0 can be determined bisection search [10]:

$$\lambda_0^o = \min_{\lambda_0\geq 0}\{\sum_{k=1}^{M}\|\mathbf{v}_k\|_2^2 \leq P_{\mathrm{m}}\}. \tag{21}$$

B. Optimize $\boldsymbol{\Theta}$

In this subsection, we consider the optimization of $\boldsymbol{\Theta}$ with $\boldsymbol{\beta}$, $\boldsymbol{\xi}$ and $\mathbf{W}$ being fixed. Specifically, we optimize $\boldsymbol{\Theta}$ by iteratively optimizing one RC at IRS k at a time while keeping the rest fixed.

First, we transform the expression of $\boldsymbol{\Theta}$ first:

$$\begin{aligned}\left|\mathbf{h}_{\mathrm{eq},k}^H\mathbf{v}_i\right|^2 &= \left|(\mathbf{f}_k+\tilde{c}_j\mathbf{G}_k^H\boldsymbol{\phi}_k^H\mathbf{h}_k)^H\mathbf{v}_i\right| = \left|B_{ki}+\tilde{c}_j^*\boldsymbol{\theta}_k^H\mathbf{p}_{ki}\right|^2 \\ &= \left|\mathbf{f}_k^H\mathbf{v}_i+\tilde{c}_j^*\alpha_k\boldsymbol{\theta}_k^H\mathrm{diag}(\mathbf{h}_k^H)\mathbf{G}_k\mathbf{v}_i\right|^2,\end{aligned} \tag{22}$$

where $B_{ki} = \mathbf{f}_k^H\mathbf{v}_i$, $\mathbf{p}_{ki} = \alpha_k\mathrm{diag}(\mathbf{h}_k^H)\mathbf{G}_k\mathbf{v}_i$, and $\boldsymbol{\theta}_k \in \mathbb{C}^{N_k\times 1}$ denotes the RCs at IRS k. Then, similar to Section III-A, we apply quadratic transform to eliminate the fractional terms, and (P1-a) can be reformulated into the following equivalent problem

$$\text{(P1-c):} \quad \max_{\boldsymbol{\Theta},\boldsymbol{\varepsilon}} \quad f_{q2}(\boldsymbol{\Theta},\boldsymbol{\varepsilon}) \quad \text{s.t.} \quad (9),$$

where $\boldsymbol{\varepsilon}$ is a set of new auxiliary variables, and the objective function of (P1-c) is defined by

$$\begin{aligned}&f_{q2}(\boldsymbol{\Theta},\boldsymbol{\varepsilon}) \\ &= \sum_{k=1}^{K}\sum_{j=1}^{2}[2\,\mathrm{Re}\{\varepsilon_{\mathrm{s},kj}^*\sqrt{\frac{\rho_{\mathrm{s},k}}{2}(1+\beta_{\mathrm{s},kj})}(B_{kk}+\tilde{c}_j^*\boldsymbol{\theta}_k^H\mathbf{p}_{kk})\} \\ &-|\varepsilon_{\mathrm{s},kj}|^2(\left|B_{kk}+\tilde{c}_j^*\boldsymbol{\theta}_k^H\mathbf{p}_{kk}\right|^2+\sigma_{\mathrm{s},kj}^2)] \\ &+\sum_{k=1}^{K}[2\,\mathrm{Re}\{\varepsilon_{\mathrm{c},k}^*\sqrt{\frac{\rho_{\mathrm{c},k}}{\Gamma}(1+\beta_{\mathrm{c},k})}\boldsymbol{\theta}_k^H\mathbf{p}_{kk}) \\ &-|\varepsilon_{\mathrm{c},k}|^2(\frac{N_T}{\Gamma}\left|\boldsymbol{\theta}_k^H\mathbf{p}_{kk}\right|^2+\sigma_{\mathrm{c},k}^2)]+const(\boldsymbol{\beta}),\end{aligned} \tag{23}$$

where

$$\sigma_{\mathrm{s},kj}^2 = \sum\nolimits_{i=1,i\neq k}^{K}\left|B_{ki}+\tilde{c}_j^*\boldsymbol{\theta}_k^H\mathbf{p}_{ki}\right|^2+\sigma_k^2, \tag{24}$$

$$\sigma_{\mathrm{c},k}^2 = \sum\nolimits_{i=1,i\neq k}^{K}|B_{ki}|^2+\sum\nolimits_{i=1,i\neq k}^{K}\left|\boldsymbol{\theta}_k^H\mathbf{p}_{ki}\right|^2+\sigma_k^2. \tag{25}$$

The optimal value of ξ_i can be easily obtained as

$$\varepsilon_{\mathrm{c},k}^o = \sqrt{\frac{\rho_{\mathrm{c},k}}{\Gamma}(1+\beta_{\mathrm{c},k})}\frac{\boldsymbol{\theta}_k^H\mathbf{p}_{kk}}{(N_T/\Gamma)\left|\boldsymbol{\theta}_k^H\mathbf{p}_{kk}\right|^2+\sigma_{\mathrm{c},k}^2}, \tag{26}$$

$$\varepsilon_{\mathrm{s},kj}^o = \sqrt{\frac{\rho_{\mathrm{s},k}}{2}(1+\beta_{\mathrm{s},kj})}\frac{B_{kk}+\tilde{c}_j^*\boldsymbol{\theta}_k^H\mathbf{p}_{kk}}{\left|B_{kk}+\tilde{c}_j^*\boldsymbol{\theta}_k^H\mathbf{p}_{kk}\right|^2+\sigma_{\mathrm{s},kj}^2}, \tag{27}$$

After simple calculating deduction, $f_{q2}(\boldsymbol{\Theta},\boldsymbol{\varepsilon})$ can be transformed to a simpler form:

$$f_{q3}(\boldsymbol{\Theta},\boldsymbol{\varepsilon}) = -\sum_{k=1}^{K}\boldsymbol{\theta}_k^H\mathbf{U}_k\boldsymbol{\theta}_k+\sum_{k=1}^{K}2\,\mathrm{Re}\{\boldsymbol{\theta}_k^H\mathbf{z}_k\}+const, \tag{28}$$

where

$$\begin{aligned}\mathbf{U}_k =& \sum_{j=1}^{2}\sum_{i=1}^{K}|\varepsilon_{\mathrm{s},kj}|^2|\tilde{c}_j|^2\mathbf{p}_{ki}\mathbf{p}_{ki}^H \\ &+|\varepsilon_{\mathrm{c},k}|^2(\frac{N_T}{\Gamma}\mathbf{p}_{kk}\mathbf{p}_{kk}^H+\sum_{i=1,i\neq k}^{K}\mathbf{p}_{ki}\mathbf{p}_{ki}^H),\end{aligned} \tag{29}$$

$$\begin{aligned}\mathbf{z}_k =& \sum_{j=1}^{2}\tilde{c}_j^*[\varepsilon_{\mathrm{s},kj}^*\sqrt{\frac{\rho_{\mathrm{s},k}}{2}(1+\beta_{\mathrm{s},kj})}\mathbf{p}_{kk} \\ &-|\varepsilon_{\mathrm{s},kj}|^2\sum_{i=1}^{K}B_{ki}^*\mathbf{p}_{ki}]+\varepsilon_{\mathrm{c},k}^*\sqrt{\frac{\rho_{\mathrm{c},k}}{\Gamma}(1+\beta_{\mathrm{c},k})}\mathbf{p}_{kk}.\end{aligned} \tag{30}$$

Then we convert the vector expressions into the scalar expressions, which are shown as

$$\sum_{k=1}^{K}\boldsymbol{\theta}_k^H\mathbf{z}_k = \theta_{kn}^*z_{kn}+\sum_{i=1,i\neq n}^{N}\theta_{ki}^*z_{ki}+\sum_{l=1,l\neq k}^{K}\sum_{i=1}^{N_k}\theta_{li}^*z_{li}, \tag{31}$$

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$$\sum_{k=1}^{K} \boldsymbol{\theta}_k^H \mathbf{U}_k \boldsymbol{\theta}_k = \theta_{kn}^* u_{kn,kn} \theta_{kn} + 2\mathrm{Re}\{ \sum_{j=1, j\neq n}^{N_k} \theta_{kn}^* u_{kn,kj} \theta_{kj} \}$$
$$+ \sum_{j=1, j\neq n}^{N_k} \sum_{i=1, i\neq n}^{N_k} \theta_{ki}^* u_{ki,kj} \theta_{kj} + \sum_{l=1, l\neq k}^{K} \sum_{j=1,}^{N_k} \sum_{i=1}^{N_k} \theta_{li}^* u_{li,lj} \theta_{lj}, \quad (32)$$

where θ_{kn} and z_{kn} denote the n-th element of vector $\boldsymbol{\theta}_k$ and $\mathbf{z}_k$, respectively, and $u_{ki,kj}$ denotes the element at i-th row and j-th column of $\mathbf{U}_k$. By substituting (31) and (32) into (28) and dropping the irrelevant constant terms, we have

$$f_{q4}(\theta_{kn}) = 2\,\mathrm{Re}\{\theta_{kn}^* A_{2,kn}\} - |\theta_{kn}|^2 A_{1,kn}, \quad (33)$$

where $A_{1,kn} = u_{kn,kn}$ and $A_{2,kn} = z_{kn} - \sum_{j=1, j\neq n}^{N_k} u_{kn,kj}\theta_{kj}$. Then (P1-c) is decomposed into $\sum_{k=1}^{K} N_k$ subproblems, and the subproblem for optimizing n-th RC at IRS k defined by

$$\text{(P1-d-}k,n\text{):} \quad \max_{\theta_{kn}} \quad f_{q4}(\theta_{kn}) \quad \text{s.t.} \quad (9).$$

It can be shown that every subproblem has a concave objective function with respect to the corresponding θ_{kn}, so the closed-form for θ_{kn} can be derived for all the three RC constraints.

- For $\theta_{kn} \in \mathcal{F}_1$: In this case, $|\theta_{kn}|^2 \leq 1$. The optimal RC can be obtained by

$$\theta_{kn}^o = \begin{cases} 0, \text{if } A_{2,kn} = 0, \\ \dfrac{A_{2,kn}}{|A_{2,kn}|}, \text{else if } A_{1,kn} = 0, \\ \dfrac{A_{2,kn}}{|A_{2,kn}|} \min\left\{1, \dfrac{|A_{2,kn}|}{A_{1,kn}}\right\}, \text{ otherwise.} \end{cases} \quad (34)$$

- For $\theta_{kn} \in \mathcal{F}_2$: In this case, $|\theta_{kn}|^2 = 1$. The optimal RC can be obtained by

$$\angle\theta_{kn}^o = \arg \min_{\varphi_{kn} \in [0, 2\pi]} |\varphi_{kn} - \angle A_{2,kn}|. \quad (35)$$

- For $\theta_{kn} \in \mathcal{F}_3$: In this case, $|\theta_{kn}|^2 = 1$. The optimal RC can be obtained by

$$\angle\theta_{kn}^o = \arg \min_{\varphi_{kn} \in \left\{0, \frac{2\pi}{\tau}, \ldots, \frac{2\pi(\tau-1)}{\tau}\right\}} |\varphi_{kn} - \angle A_{2,kn}|. \quad (36)$$

C. Algorithm Development

The proposed algorithm is summarized in **Algorithm 1** and described as follows. First, $\mathbf{W}^{(0)}$ and $\boldsymbol{\Theta}^{(0)}$ are initialized with some feasible values. Then the value of $\mathbf{W}^{(0)}$ and $\boldsymbol{\Theta}^{(0)}$ are updated iteratively. In the i-th iteration, with the value of $\boldsymbol{\Theta}^{(i-1)}$ obtained in the previous iteration, we will obtain the auxiliary variable $\boldsymbol{\beta}^{(i)}$ according to (12) and (13) and the auxiliary variable $\boldsymbol{\xi}^{(i)}$ according to (16) and (17), and then we obtain $\mathbf{W}^{(i)}$ according to (18). Next, with the updated value of $\mathbf{W}^{(i)}$, $\boldsymbol{\beta}^{(i)}$ and $\boldsymbol{\xi}^{(i)}$, we will obtain the auxiliary variable $\boldsymbol{\varepsilon}^{(i)}$ according to (26) and (27) first, and then obtain $\boldsymbol{\Theta}^{(i)}$ according to (34), (35) or (36). This process will be repeated until the objective function of (P1-a) $f_r(\mathbf{W}, \boldsymbol{\Theta}, \boldsymbol{\beta})$ converges.

Algorithm 1 The iterative method for solving (P1).

Step 0:Initialize $\mathbf{W}$ and $\boldsymbol{\Theta}$ with feasible values
Repeat
  Step 1:Update $\boldsymbol{\beta}$ with (12) and (13)
  Step 2:Update $\boldsymbol{\xi}$ with (16) and (17)
  Step 3:Update $\mathbf{W}$ with (18)
  Step 4:Update $\boldsymbol{\varepsilon}$ with (26) and (27)
  Step 5:Update $\boldsymbol{\Theta}$ with (34), (35) or (36)
until the value of function $f_r(\mathbf{W}, \boldsymbol{\Theta}, \boldsymbol{\beta})$ in (11) converges

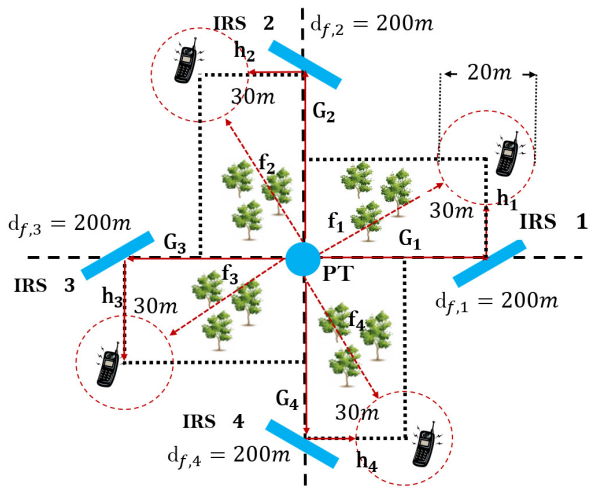

Fig. 2: The simulated scenario of the MU-MISO SR system

IV. Simulation

In this subsection, we investigate the rate performance of the proposed system by numerical simulations. Similar to [9], we consider an IRS-aided femtocell network illustrated in Fig. 2, in which the PT is located at (0,0). We set $K = 4$ and $M = 4$, and the IRSs are located at (200m,0), (0,200m), (-200m,0), and (0,-200m), respectively. Each PR is randomly distributed in a circle with diameter 20m. As shown in Fig. 2, these circles are cented at (200m,30m), (-30m,200m), (-200m,-30m), and (30m,-200m), respectively. Besides, $\mathbf{f}_k$ is assumed to follow the Rayleigh fading, and the large-scale fading of $\mathbf{f}_k$ is denoted by $\kappa_{f,k} = \beta_{f,k} d_{f,k}^{-\varrho_f}$, where $\beta_{f,k}$= -30dB denotes the path loss at the reference distance, $d_{f,k}$ denotes the link distance between PT and PR k, and ϱ_f = 3.5 denotes the path loss exponent of $\mathbf{f}_k$. On the other hand, the large scale fading of $\mathbf{G}_k$ and $\mathbf{h}_k$ are denoted by $\kappa_{G,k} = \beta_{G,k} d_{G,k}^{-\varrho_G}$ and $\kappa_{h,k} = \beta_{h,k} d_{h,k}^{-\varrho_h}$, respectively, and they are assumed to follow the Rician fading since the locations of IRSs are usually carefully chosen [9]. Hence, we have $\varrho_h = \varrho_G = 2$ and the Rician factors equal to 10. Besides, we set $\beta_{h,k} = \beta_{G,k}$=-30dB. For simplicity, we set the size of each IRS as 100, the reflection gain of each IRS as 10 dB, the background noise at each PR as -118 dBm, the SINR gap Γ as 1, the weighted factors ρ_{sk} and ρ_{ck} equal to each other and they are subject to $\sum_{k=1}^{K} (\rho_{sk} + \rho_{ck}) = 1$.

Fig. 3 illustrates the sum rate of the primary and IoT transmissions (R_1) for different schemes with respect to the

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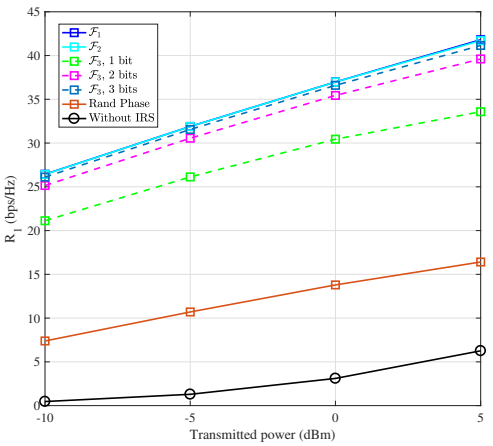

Fig. 3: R_1 vs P_m: N=100

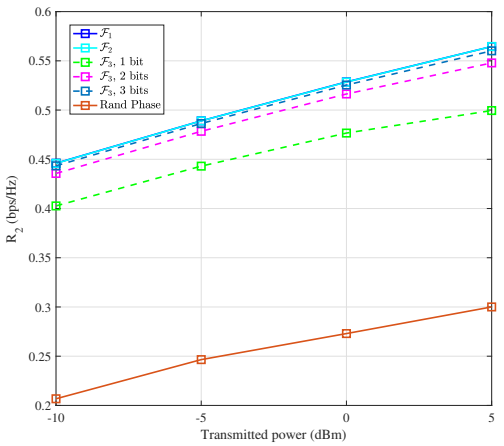

Fig. 4: R_2 vs P_m: N=100

transmit power of PT (P_m). It is seen that as compared to the case of without IRS, significant performance gains are achieved when the optimal phase constraint $\mathcal{F}_1$ is considered, and the performance loss is negligible when $\theta \in \mathcal{F}_2$, which reveals that the amplitude of the optimal θ is still very close to 1 when $\theta \in \mathcal{F}_1$. In addition, we observe that the discrete phase-shifters still achieve significant performance gains: '3-bit' phase-shifter may obtain almost the full beamforming gain compared with the optimal one, and even '1-bit' shifter achieves more than 20 bps/Hz gain contrasting with the baseline system without IRS. Besides, the performance gain of the random passive beamforming scheme is still relatively high, achieving at 9 bps/Hz when P_m= -5dBm. However, the quantization losses of the discrete phase-shifters increase as P_m increases. Hence, we prefer high-order quantization for the cases with large P_m. Next, in Fig. 4, we illustrate the sum rate of IoT transmissions (R_2) for different schemes. It is seen that we get the results similar to Fig. 3 as expected, and even '1-bit' phase-shifter enables more than 0.4 bps/Hz IoT transmissions when P_m = -10 dBm, which verifies that IRSs can simultaneously assist and communicate with PRs in the proposed system.

V. Conclusion

In this paper, we have proposed an IRS enhanced MU-MISO SR system, where each RIS acting as an IoT device transmits information to a corresponding PR. We have formulated the problem of maximizing the WSR of both primary and IoT transmissions. To tackle the non-convex problem, an algorithm based on FP and AO method has been proposed. Numerical results have shown the proposed system achieves a significant performance gain over different benchmark schemes.

Acknowledgement

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